

Study Student Shop (SSS)

Comprehensive Product Requirements Document (PRD) & Technical Blueprint v2.4

Document Version: 2.4.0	Target Release: Q4 2026
Product Status: Ready for Engineering / Production	Target Market: Indian Higher Education & Global Students
Authors: SSS Core Architecture & Product Team	Confidentiality: Tier-1 Internal / Stakeholders

1. Executive Summary & Product Vision

Study Student Shop (SSS) is a unified, smart, all-in-one student commerce and academic utility ecosystem. Historically, college students across India and worldwide face severe friction when sourcing semester textbooks, calculating university-specific grades with irregular conversion formulas, liquidating used course materials, and hunting for the lowest-priced new textbooks across fragmented e-commerce platforms. SSS resolves this by merging a peer-to-peer (P2P) resale and exchange marketplace, a real-time multi-platform price aggregator (Amazon, Flipkart, Bookswagon, Google Books), an automated lowest-price dropshipping/auto-order fulfillment bot, an e-book digital library, and an official Indian Universities CGPA/Marks engine (supporting AKTU, VTU, SPPU, Mumbai Univ, DU, Anna Univ, MAKAUT, JNTU, GTU, etc.) under an enterprise-grade administrative governance framework.

2. Strategic Value Propositions & Target Personas

Stakeholder / Persona	Core Pain Points	SSS Strategic Solution
College Student (Buyer/Seller)	High cost of new books, unorganized campus seniors resale, uncertainty about edition compatibility.	Campus P2P resale/exchange portal with condition ratings, instant ISBN scan, and campus handover.
Bargain Seeker & Researcher	Hours wasted comparing Amazon vs Flipkart vs Bookswagon vs Local stores for cheapest price.	One-click live search with automated price comparisons, lowest price badge, and single-click automated order execution.
Engineering/Medical/Arts Student	Confusing CGPA-to-percentage conversion rules across different Indian universities (e.g. (CGPA-0.75)*10 vs 9.5*CGPA).	Automated University CGPA/SGPA engine with syllabus credit matrices, marks prediction, and downloadable marksheets.
Platform Administrator	Downtime management, content moderation, fraud prevention, tracking dropship margins.	Live Admin Command Center with Instant Maintenance Toggle, User RBAC, Real-time Scraper Logs, and Dropship monitor.

3. Detailed Functional Modules & Feature Requirements

3.1 Peer-to-Peer (P2P) Book Resale & Book Exchange

- Book Resale Listing:** Students can list used textbooks specifying Title, Author, Edition, Subject, University/Branch, Condition (Like New, Good, Fair, Acceptable), Price, Original MRP, and campus handover location.
- Book Exchange (Barter/Swap):** Students can mark books as 'Available for Exchange' and specify target books or subjects they need in return.
- Exchange Proposal Workflow:** Student B sends an exchange request proposing one of their listed books. Student A can Accept, Decline, or Propose Counter-offer with automated chat and campus rendezvous setup.
- Safety & Condition Badges:** Verified student badge, book condition photo gallery, condition verification checklist upon handover.

3.2 Real-Time Google Books & Multi-Vendor Price Aggregator

- **Ultra-Fast Live Search:** Debounced real-time search queries connecting to Google Books API + SSS Local Cache. Instant auto-complete with thumbnails, ISBN-10/13, publisher, author, and edition metadata.
- **Multi-Platform Price Intelligence:** Scrapes/aggregates live price and in-stock status across Amazon.in, Flipkart, Bookswagon, and SSS Verified Used Sellers.
- **Lowest Price Highlighting:** Visually pinpoints the lowest-cost vendor with price differentials, estimated delivery days, and savings badges.

3.3 Automated Dropshipping / Arbitrage Auto-Ordering Engine

- **Unified SSS Checkout:** User makes payment directly on SSS (Cards, UPI, NetBanking, Campus Pay).
- **Automated Order Bot:** An asynchronous backend worker receives the order, identifies the external platform currently offering the lowest price (e.g. Amazon at ₹320 vs Flipkart at ₹399), logs in via platform merchant/affiliate bot or API, places the order with the customer's shipping address, and captures external order ID.
- **Order Lifecycle Management:** User receives unified real-time tracking: [Payment Confirmed] → [Auto-Ordered at Lowest Vendor] → [Dispatched] → [Out for Delivery] → [Delivered].
- **Failsafe Fallback:** In case of vendor stockouts or price spikes, the auto-order queue alerts admin and switches to 2nd lowest vendor or initiates instant refund.

3.4 E-Books & Digital Library Suite

- **Digital Catalog:** Instant access to academic e-books, lecture notes, lab manuals, and previous years' question papers (PYQs).
- **Interactive Web E-Reader:** In-browser reading interface with zoom, full-screen, page jump, night/sepia mode, and bookmarking.
- **Instant Purchase & Download:** E-books can be unlocked via instant micro-transactions and downloaded as DRM-stamped PDFs.

3.5 Indian Universities Academic Engine (CGPA, SGPA, Marks)

- **Multi-University Grading Schemes:** Supports distinct grading formulas for major Indian Universities:

University / Board	Grading System / Formula	Percentage Conversion Formula
AKTU (Lucknow)	10-Point Absolute Grading (O, A+, A, B+, B, C, P, F)	Percentage = (CGPA - 0.75) * 10
VTU (Karnataka)	10-Point CBCS Grade Points (S, A, B, C, D, E, F)	Percentage = (CGPA - 0.75) * 10 (Pre-2022) / CGPA * 10 (2022+)
Delhi University (DU)	UGC-CBCS 10-Point Letter Scale	Percentage = CGPA * 9.5
Mumbai University (MU)	7-Point & 10-Point CBSGS Scale	Percentage = 7.1 + 0.1 * (CGPA - 7) * 10 (for >=7.0) / CGPA * 7.25
SPPU (Pune Univ)	10-Point Credit System (O, A+, A, B+, B, C, D, F)	Percentage = (CGPA * 10) - 7.5 (for >= 7.0) or slab conversion
Anna University	10-Point Scale (O, A+, A, B+, B, RA)	Percentage = CGPA * 10
MAKAUT (WBUT)	10-Point CBCS Engineering Scale	Percentage = (CGPA - 0.75) * 10
JNTU (Hyderabad/Kakinada)	10-Point CBCS Scale	Percentage = (CGPA - 0.5) * 10
GTU (Gujarat)	SPI / CPI 10-Point Scale	Percentage = (CPI - 0.5) * 10

- **Internal Marks & Target CGPA Predictor:** Calculates how many marks in End-Semester exams are required based on Mid-Term/Internal marks to achieve student's goal (e.g., 8.5 CGPA).
- **Exportable Marksheet PDF:** One-click generation of student SGPA/CGPA transcript with university header.

3.6 User Profile & Student Hub

- **Active Resale Listings:** Manage prices, mark items as sold, toggle visibility.
- **Exchange Dashboard:** Inbox for trade offers with accept/reject/chat modals.
- **Order & Shipment History:** Real-time dropship order progression with tracking IDs.
- **Academic Vault:** Saved semester grade reports and target CGPA milestones.

3.7 Admin Command Center & Governance

- **Site Maintenance Toggle:** Instantly place the application in Maintenance Mode with custom banner, estimated resolution ETA, and admin bypass authentication.
- **User & Role Management (RBAC):** Add, edit, ban, delete users, and assign granular roles (Admin, Moderator, Student, Verified Seller).
- **Live Audit Logs:** Detailed audit trail logging logins, role grants, maintenance state changes, auto-order dropship dispatches, and pricing scrapers.
- **Arbitrage & Auto-Order Monitor:** Live status of external vendor orders, margin tracker, and manual retry override.

4. Technical & System Architecture

The SSS platform employs a modern modular, high-throughput full-stack architecture built on Next.js 14/15, Tailwind CSS, Express/Node.js API services, SQLite/PostgreSQL with Prisma ORM, Redis for live caching, and an asynchronous worker pipeline for the dropshipping auto-order execution engine.

Layer	Technology Stack	Key Responsibilities
Frontend Presentation	Next.js / React, Tailwind CSS, Lucide Icons, Framer Motion, Canvas Confetti	Responsive SPA/SSR, real-time debounced search, e-reader, academic calculators, and interactive admin control panel.
Application & API	Node.js / Express REST & Next.js API Routes	Authentication (JWT), P2P matchmaking, maintenance guard middleware, order execution, university calculation algorithms.
Data & Persistence	Prisma ORM with SQLite / PostgreSQL, In-Memory Store	User profiles, book listings, exchange proposals, order transactions, audit logs, and university syllabus templates.
External Integrations	Google Books API, E-commerce Price Scrapers / Adapters	Live book metadata lookup, real-time price monitoring across Amazon, Flipkart, Bookswagon.
Worker & Automation	Async Auto-Order Queue / Simulated Dropship Bot	Evaluates lowest priced vendor, auto-places purchase on behalf of user, generates tracking number and updates order state.

5. Non-Functional Requirements (NFR) & Security

- **Performance:** Live book search response time < 150ms via debounced Google Books API + local caching.
- **High Availability & Maintenance Guard:** Graceful maintenance mode with 503 HTTP status code handling and zero data leakage.
- **Security:** Role-based access control (RBAC), bcrypt password hashing, sanitized inputs against XSS/SQLi.
- **Auditability:** Comprehensive immutable logging for all administrative modifications, user permission updates, and order dropship requests.

6. Release Roadmap & Success Metrics

- **Milestone 1 (MVP - Current):** P2P Resell/Exchange, Live Price Aggregator, Auto-Order Dropship Engine, Academic Calculators, E-Reader, Maintenance Guard, Admin Dashboard.

- **Milestone 2:** Direct WhatsApp/SMS notifications for exchange requests and dropship delivery tracking.
- **Milestone 3:** AI-driven book condition grading from camera scans and personalized course syllabus book recommendations.
- **Success Metrics (KPIs):** Average savings per student (> 40% vs retail MRP), auto-order dropship success rate (> 99.2%), platform daily active users (DAU), CGPA calculation accuracy (100%).

End of Product Requirements Document (PRD) — Study Student Shop (SSS)